DEPARTMENT OF COMPUTER SCIENCE
UNIVERSITY COLLEGE OF ENGINEERING (A)
Osmania University, Hyderabad-07

CLASS TEST-I

Subject : Number Theory and Cryptography

Class: B.E(CSE)
Semester: V
Academic Year: 2025-2026

Max Marks: 30
Duration: 1 hr ½
Date : 08-09-2026

		Marks
	PART A **Answer All Questions (5 x 2 = 10 M)**	
1. a)	Use the additive cipher with key = 15 to encrypt the message "hello".	2
b)	What are the primitive roots of Z_7 ?	2
c)	What is Cryptography?	2
d)	What is symmetric key – encryption?	2
e)	How many exclusive-or operations are used in the DES cipher?	2
	PART B **Answer Any Two Questions (2 x 10 = 20 M)**	
1.	a) Explain the Chinese Remainder Theorem.Solve $x \equiv 2 \pmod 3$, $x \equiv 3 \pmod 5$, $x \equiv 2 \pmod 7$	5
	b)Eve has intercepted the cipher text " UVACLYFZLJBYL". Show how she can use a brute-force attack to break the cipher.	5
2.	a) Distinguish between passive and active security attacks with suitable examples.	5
	b) In RSA: Given p = 19, q = 23, and e = 3, find n, φ(n), and d. M=5,C=?.	5
3.	Explain DES algorithm. Show the result of passing 110111 through S-box 3. S-box 3 01 2 34 5 6 7 8 9 10 11 12 13 14 15 0 10 00 09 14 06 03 15 05 01 13 12 07 11 04 02 08 1 13 07 00 09 03 04 06 10 02 08 05 14 12 11 15 01 2 13 06 04 09 08 15 03 00 11 01 02 12 05 10 14 07 3 01 10 13 00 06 09 08 07 04 15 14 03 11 05 02 12	10